Lab Assignment 1:DBMS (ABC Information Technology)

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1.1: Database Schema Creation

Question 1: Departments Table

Create a table named **DEPARTMENTS** consisting of columns DEPT_NAME, DEPT_NUMBER, MANAGER_ID, and MANAGER_START_DATE. Set DEPT_NUMBER as the primary key.

SQL Query:

```
SQL
```

```
CREATE TABLE DEPARTMENTS (
  DEPT_NAME VARCHAR(50),
  DEPT_NUMBER INT PRIMARY KEY,
  MANAGER_ID INT,
  MANAGER_START_DATE DATE
);
```

Schema Structure:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
DEPT_NAME	varchar(50)	YES		NULL	
DEPT_NUMBER	int	NO	PRI	NULL	
MANAGER_ID	int	YES		NULL	
MANAGER_START_DATE	date	YES		NULL	

Question 2: Projects Table

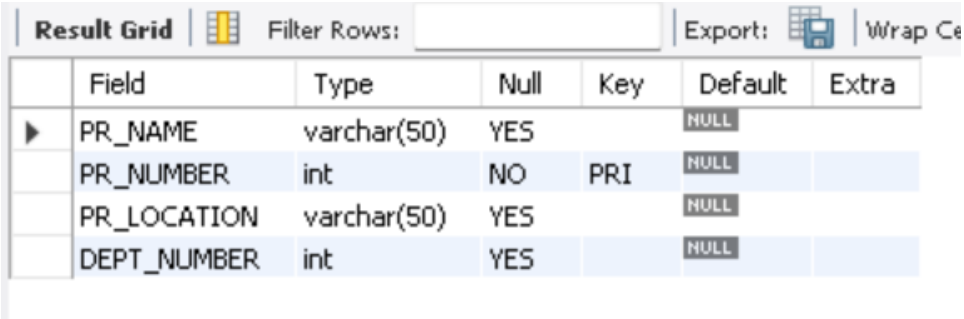
Create a table named **PROJECTS** consisting of columns PR_NAME, PR_NUMBER, PR_LOCATION, and DEPT_NUMBER. Set PR_NUMBER as the primary key.

SQL Query:

```
SQL
```

```
CREATE TABLE PROJECTS (
  PR_NAME VARCHAR(50),
  PR_NUMBER INT PRIMARY KEY,
  PR_LOCATION VARCHAR(50),
  DEPT_NUMBER INT );
```

Schema Structure:


Result Grid | Filter Rows: | Export: | Wrap Ce

Field	Type	Null	Key	Default	Extra
PR_NAME	varchar(50)	YES		NULL	
PR_NUMBER	int	NO	PRI	NULL	
PR_LOCATION	varchar(50)	YES		NULL	
DEPT_NUMBER	int	YES		NULL	

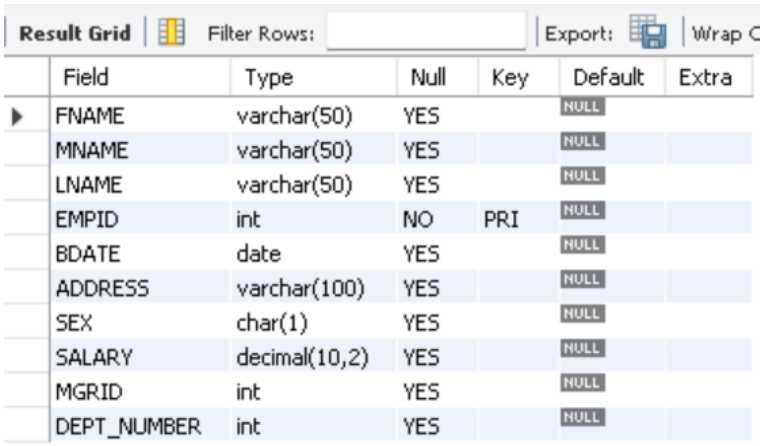
Question 3: Employees Table

Create a table named **EMPLOYEES** consisting of columns FNAME, MNAME, LNAME, EMPID, BDATE, ADDRESS, SEX, SALARY, MGRID, and DEPT_NUMBER. Set EMPID as the primary key.

SQL Query:

SQL

```
CREATE TABLE EMPLOYEES (
  FNAME VARCHAR(50),
  MNAME VARCHAR(50),
  LNAME VARCHAR(50),
  EMPID INT PRIMARY KEY,
  BDATE DATE,
  ADDRESS VARCHAR(100),
  SEX CHAR(1),
  SALARY DECIMAL(10,2),
  MGRID INT,
  DEPT_NUMBER INT
);
```


Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
FNAME	varchar(50)	YES		NULL	
MNAME	varchar(50)	YES		NULL	
LNAME	varchar(50)	YES		NULL	
EMPID	int	NO	PRI	NULL	
BDATE	date	YES		NULL	
ADDRESS	varchar(100)	YES		NULL	
SEX	char(1)	YES		NULL	
SALARY	decimal(10,2)	YES		NULL	
MGRID	int	YES		NULL	
DEPT_NUMBER	int	YES		NULL	

Question 4: Dependents Table

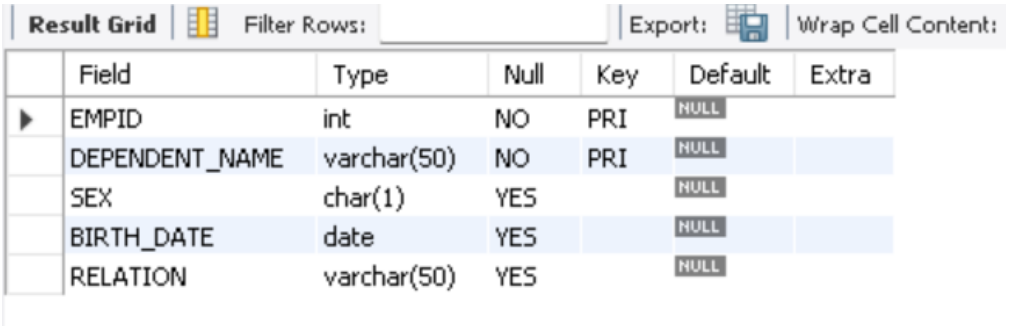
Create a table named **DEPENDENTS** consisting of columns EMPID, DEPENDENT_NAME, SEX, BIRTH_DATE, and RELATION. Define a composite primary key using EMPID and DEPENDENT_NAME.

SQL Query:

```
SQL

CREATE TABLE DEPENDENTS (
  EMPID INT,
  DEPENDENT_NAME VARCHAR(50),
  SEX CHAR(1),
  BIRTH_DATE DATE,
  RELATION VARCHAR(50),
  PRIMARY KEY (EMPID, DEPENDENT_NAME)
);
```

Schema Structure:



Field	Type	Null	Key	Default	Extra
EMPID	int	NO	PRI	NULL	
DEPENDENT_NAME	varchar(50)	NO	PRI	NULL	
SEX	char(1)	YES		NULL	
BIRTH_DATE	date	YES		NULL	
RELATION	varchar(50)	YES		NULL	

Question 5: Works_On Table

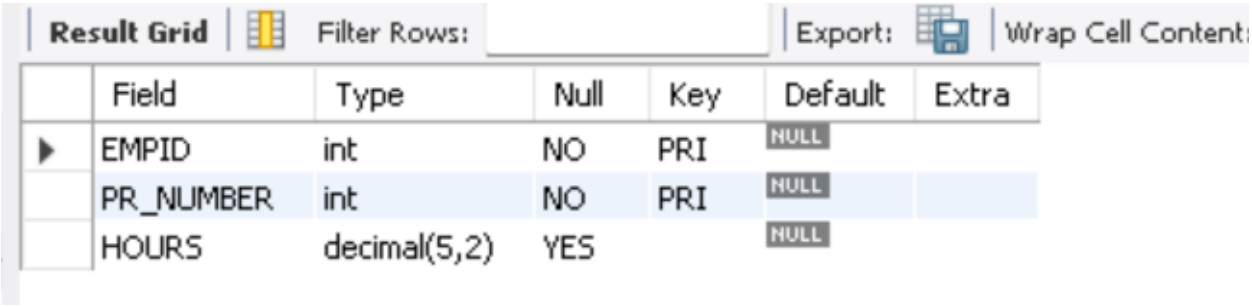
Create a table named **WORKS_ON** consisting of columns EMPID, PR_NUMBER, and HOURS. Define a composite primary key using EMPID and PR_NUMBER.

SQL Query:

```
SQL

CREATE TABLE WORKS_ON (
  EMPID INT,
  PR_NUMBER INT,
  HOURS DECIMAL(5,2),
  PRIMARY KEY (EMPID, PR_NUMBER)
);
```

Schema Structure:



	Field	Type	Null	Key	Default	Extra
▶	EMPID	int	NO	PRI	NULL	
	PR_NUMBER	int	NO	PRI	NULL	
	HOURS	decimal(5,2)	YES		NULL	

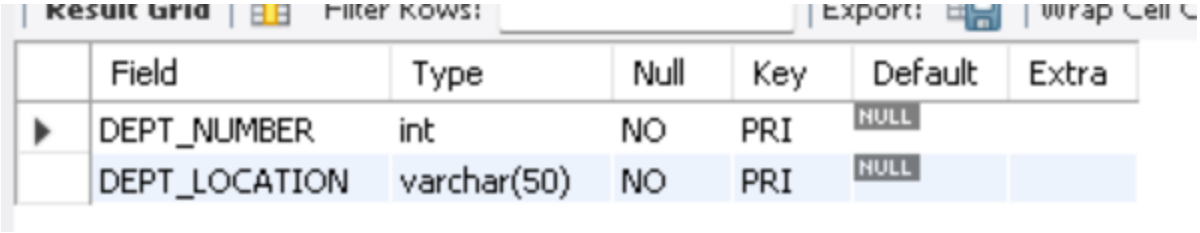
Question 6: Dept_Locations Table

Create a table named **DEPT_LOCATIONS** consisting of columns DEPT_NUMBER and DEPT_LOCATION. Define a composite primary key using DEPT_NUMBER and DEPT_LOCATION.

SQL Query:

```SQL
CREATE TABLE DEPT_LOCATIONS (
  DEPT_NUMBER INT,
  DEPT_LOCATION VARCHAR(50),
  PRIMARY KEY (DEPT_NUMBER, DEPT_LOCATION)
);
```

Schema Structure:



	Field	Type	Null	Key	Default	Extra
▶	DEPT_NUMBER	int	NO	PRI	NULL	
	DEPT_LOCATION	varchar(50)	NO	PRI	NULL	

1.2: Data Insertion

Question 1: Insert data into table DEPARTMENTS

```SQL
INSERT INTO DEPARTMENTS VALUES ('HR', 1, 101, DATE '2020-01-15');
INSERT INTO DEPARTMENTS VALUES ('Information Technology', 2, 102, DATE '2021-03-10');
INSERT INTO DEPARTMENTS VALUES ('FINANCE', 3, 103, DATE '2019-07-20');
INSERT INTO DEPARTMENTS VALUES ('MARKETING', 4, 104, DATE '2022-05-01');
```

Question 2: Insert data into table PROJECTS

SQL

```
INSERT INTO PROJECTS VALUES ('Website Dev', 201, 'Delhi', 2);
INSERT INTO PROJECTS VALUES ('App Development', 202, 'Mumbai', 2);
INSERT INTO PROJECTS VALUES ('Audit System', 203, 'Pune', 3);
INSERT INTO PROJECTS VALUES ('HR Portal', 204, 'Delhi', 1);
```

Question 3: Insert data into table EMPLOYEES

SQL

```
INSERT INTO EMPLOYEES VALUES ('Amit', 'K', 'Sharma', 101, DATE '1985-05-10', 'Delhi', 'M', 60000, NULL,
1);
INSERT INTO EMPLOYEES VALUES ('Neha', 'R', 'Verma', 102, DATE '1990-08-22', 'Mumbai', 'F', 75000, 101,
2);
INSERT INTO EMPLOYEES VALUES ('Raj', 'S', 'Mehta', 103, DATE '1988-12-05', 'Pune', 'M', 70000, 101, 3);
INSERT INTO EMPLOYEES VALUES ('Simran', 'T', 'Kaur', 104, DATE '1995-02-18', 'Chandigarh', 'F', 50000,
102, 2);
```

Question 4: Insert data into table DEPENDENTS

SQL

```
INSERT INTO DEPENDENTS VALUES (101, 'Riya', 'F', DATE '2010-06-15', 'Daughter');
INSERT INTO DEPENDENTS VALUES (102, 'Arjun', 'M', DATE '2012-09-10', 'Son');
INSERT INTO DEPENDENTS VALUES (103, 'Meera', 'F', DATE '2015-01-20', 'Daughter');
INSERT INTO DEPENDENTS VALUES (104, 'Karan', 'M', DATE '2018-03-25', 'Son');
```

Question 5: Insert data into table WORKS_ON

SQL

```
INSERT INTO WORKS_ON VALUES (101, 201, 10);
INSERT INTO WORKS_ON VALUES (102, 201, 20);
INSERT INTO WORKS_ON VALUES (103, 203, 15);
INSERT INTO WORKS_ON VALUES (104, 202, 25);
```

Question 6: Insert data into table DEPT_LOCATIONS

SQL

```
INSERT INTO DEPT_LOCATIONS VALUES (1, 'Delhi');
INSERT INTO DEPT_LOCATIONS VALUES (2, 'Mumbai');
INSERT INTO DEPT_LOCATIONS VALUES (3, 'Pune');
INSERT INTO DEPT_LOCATIONS VALUES (4, 'Bangalore');
```